

Machine Learning Term Project

Home Network Intrusion Detection

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1. Overview: Home Network Intrusion Detection?

- **Home Network:**

- WIFI router
- Laptop
- Smartphone
- IoT device

- **Intrusion Detection:**

- Detect malicious traffic
- Identify suspicious network behavior
- Protect privacy and security

1. Overview

- **Goal:**

- Detect malicious network traffic
- Binary classification:
 - Normal traffic
 - Attack traffic

- **Why Important?**

- Home networks contain personal devices
- Attacks may cause privacy leakage
- Detecting attacks is important

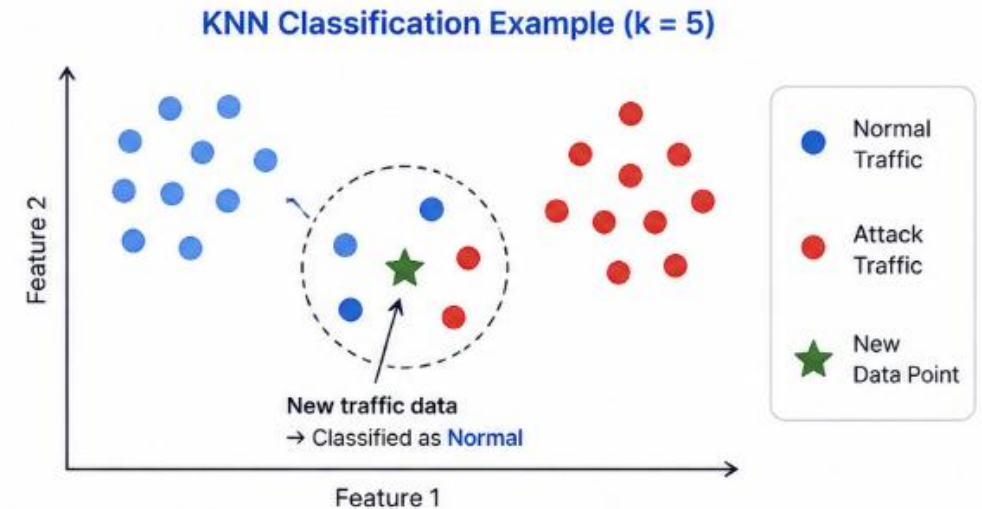
2. Dataset

- **Dataset Information**

- CICIDS2017 Dataset
 - Network Traffic Dataset
 - Preprocessed Version
-
- The dataset contains network traffic features such as packet length and port information

3. Baseline Model (KNN)

- **K-Nearest Neighbors:**
 - Distance-based classification model
 - Compares similar traffic patterns
 - Predicts labels using nearest neighbors
- **Strength**
 - Simple and intuitive model
 - Easy to implement
- **Limitation**
 - Difficult to handle complex traffic patterns
 - Sensitive to high-dimensional data



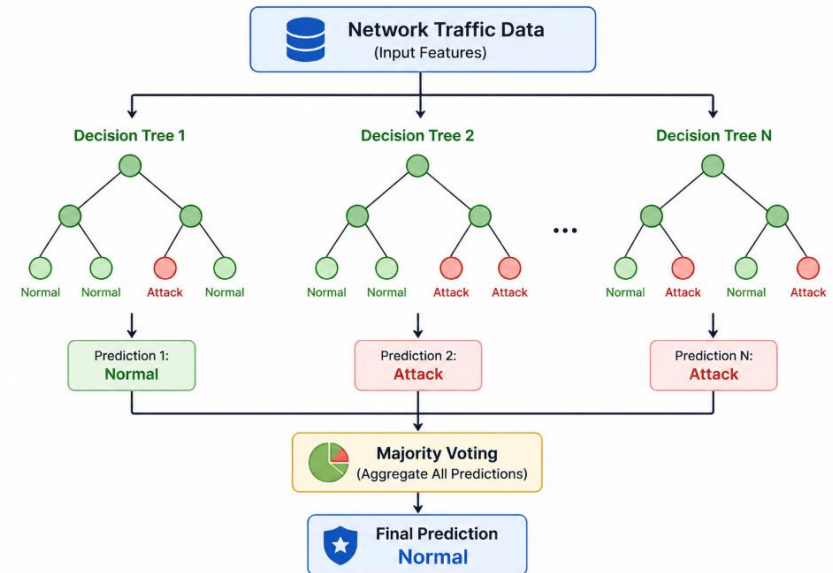
4. Main model (Random Forest)

- **Random Forest**

- Ensemble-based machine learning model
- Combines multiple decision trees

- **Strength**

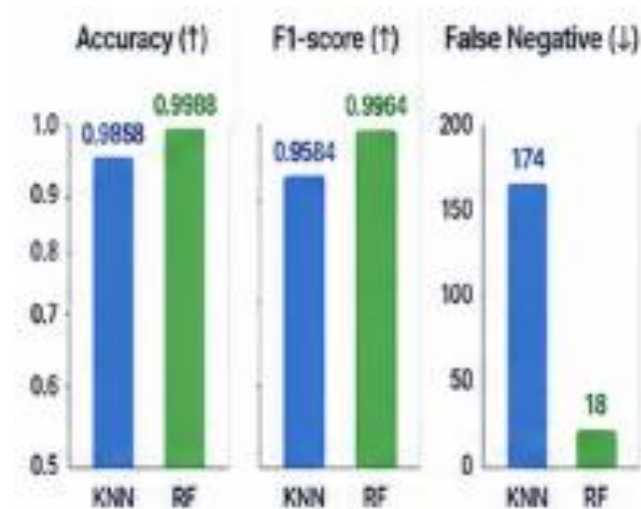
- Robust against noisy data
- Stable prediction performance
- Handles complex traffic patterns



5. Final Results

- Performance Comparision

Model	Accuracy (↑)	F1-score (↑)	False Negative (FN) (↓)
KNN (Baseline)	0.9858	0.9584	174
Random Forest (Main Model)	0.9888	0.9964	18



False Negative (Missed Attack): 174 -> 18

- Accuracy = Correct Predictions / Total Predictions
- F1-score = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
- FN : Actual Attack -> Predicted Normal

5. Final Results

- **Confusion Matrix Analysis**

	Normal	Attack
Normal	56,695 (TN)	831 (FP)
Attack	18 (FN)	8,137 (TP)

- **Interpretation:**

- Most normal traffic was correctly classified
- Most attack traffic was successfully classified
- Very small number of missed attacks
- Stable and reliable prediction performance

Strong attack detection performance with very few missed attacks

- TN: Normal traffic -> Normal traffic
- TP: Attack traffic -> Attack traffic
- FP: Normal traffic -> Attack traffic
- FN: Attack traffic -> Normal traffic

Thank You